

DISCUSSION AND REFLECTION

1.Network structure profoundly influences social outcomes by dictating the pathways and efficiency of flow (information, influence, or resources).

- Centrality: Highly central nodes (e.g., high degree, betweenness, or eigenvector centrality) are more influential.² They are often the first to adopt new behaviours or information, acting as hubs for rapid spread.
- Homophily and Segregation: Networks often exhibit homophily (the tendency for nodes to connect with similar nodes), creating tight-knit clusters or communities. This structure reinforces existing beliefs and behaviours within the cluster, leading to echo chambers and slowing the spread of novel information *between* groups.
- Tie Strength and Bridging:
 - Strong ties (close friends) facilitate the exchange of rich, complex information and are crucial for behavioural conformity and trust.
 - Weak ties (acquaintances) act as bridges between different social clusters (structural holes). According to Gravette’s "Strength of Weak Ties" theory, weak ties are the primary means by which novel, non-redundant information enters a community.
- Small-World Networks: Networks characterized by high clustering (like regular networks) but short path lengths (like random networks) allow for both local reinforcement and long-distance rapid diffusion. This structure accelerates the spread of simple contagion (like an epidemic or a simple piece of information) globally.

2.Network density is the ratio of actual connections to all possible connections, its effect on collaboration and innovation is often curvilinear that is too high or too low can be detrimental, but for different reasons.

Density Level	Collaboration & Knowledge Flow	Innovation Impact
High Density	Pros: Rapid and reliable information flow, high trust, strong consensus, easy coordination.	Exploitative Innovation: Excellent for refining and improving existing knowledge and processes (e.g., maximizing current products). However, it can lead to redundancy of information and groupthink, limiting new ideas.
Low Density	Pros: Access to diverse, non-redundant knowledge from outside the core group. Cons: Slower communication, lower trust, difficulty coordinating.	Exploratory Innovation: Essential for discovering and generating new ideas or radical breakthroughs, as nodes act as brokers of novel information from different sectors. However, if too low, it can hinder knowledge sharing and lead to fragmentation.

The Ideal Balance often involves a structure where dense clusters (for trust and efficient internal collaboration) are connected by weak ties or structural bridges (for importing new, diverse knowledge essential for innovation).

3.Graph directionality fundamentally changes how network metrics are calculated and interpreted, as it distinguishes between a relationship being mutual (undirected) or one-way (directed).

Feature	Facebook (Undirected Graph)	Twitter (Directed Graph)
Relationship	Mutual: Friendship is typically reciprocal (A is a friend of B, and B is a friend of A).	One-Way: Following is non-reciprocal (A follows B, but B does not necessarily follow A).
Key Centrality Metrics	Degree Centrality: A single count of total friends.	In-Degree Centrality: Number of followers (a measure of popularity/prestige or attractiveness).
		Out-Degree Centrality: Number of accounts followed (a measure of expansiveness or engagement).
Analysis Alteration	Focuses on shared context, reciprocated influence, and tight-knit communities (e.g., how common interests are reinforced).	Focuses on influence, reach, and information cascades (e.g., how a public figure's tweet spreads, or which accounts are influential vs. which are merely active).
Information Flow	Information flows bidirectionally, often leading to faster consensus within a closed group.	Information flow is asymmetric; high-in-degree nodes become broadcasters, controlling the initial source of spread.

In a directed graph like Twitter, one must analyse both in-degree and out-degree to understand a user's role. A user with high out-degree is an *information consumer*, while a user with high in-degree is an *information source*.

4. Analysing social networks, especially using public data from platforms, raises several critical ethical issues related to privacy and informed consent.

Privacy and Anonymity

- **Re-identification Risk:** Even if data is technically anonymized by removing names, an individual's unique pattern of network ties (their position in the graph) can often be used to re-identify them, especially when combined with other publicly available information.
- **Privacy of Non-Consenting Parties:** In traditional social network surveys, people are asked about their connections. While the respondent consents, the named contacts often do not. The researcher is collecting data about non-participants without their knowledge.
- **Inferred Sensitive Data:** Analysis can infer sensitive attributes (like political affiliation, sexual orientation, or vulnerability) based on a person's connections, even if that information wasn't explicitly shared. This is known as privacy by association.

Consent and Public Data

- **Informed Consent Ambiguity:** Most social media data is considered "public" by platform terms of service. However, users often lack the reasonable expectation of privacy that their "public" posts will be used for academic or commercial analysis without their explicit knowledge.
- **Comprehension of Consent:** Users clicking "I Agree" to long, complex terms of service rarely constitute truly informed consent regarding social network analysis methods, such as inferring their personality or influence.